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Degree: B.Tech

Roll no.:.....  
Semester: 3

Mid Semester Examination, September 2025

Course Code: ITMTC04

Course title: Optimization Principles and Techniques

Time: 1 hour 30 minutes

Maximum Marks. 25

Note: Attempt all the questions in the given order only. Missing data/information(if any), may be suitably assumed and mentioned in the answer.

| Q. No. | Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks | CO  |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|
| 1a     | Solve the following LPP using graphical method:<br>Max $Z = 2x_1 + 3x_2$<br>subject to<br>$x_1 + x_2 \leq 30$ ,<br>$x_1 - x_2 \geq 0$ ,<br>$x_2 \geq 3$ ,<br>$0 \leq x_1 \leq 20$ ,<br>$0 \leq x_2 \leq 12$ .                                                                                                                                                                                                                                                                                                                     | 2.5   | CO1 |
| 1b     | Solve the following linear programming problem.<br>Max $Z = x + 3y + 6z + w$<br>subject to<br>$5x + y + 6z + 7w \leq 20$ ,<br>$6x + 2y + 2z + 9w \geq 40$ ,<br>$x, y, z, w \geq 0$ .                                                                                                                                                                                                                                                                                                                                              | 2.5   | CO1 |
| 2a     | Find all degenerate basic feasible solutions of the system<br>$x_1 + x_2 + x_3 = 3$ ,<br>$x_1 - x_2 + x_4 = 0$ ,<br>$x_1, x_2, x_3, x_4 \geq 0$ .                                                                                                                                                                                                                                                                                                                                                                                 | 2.5   | CO1 |
| 2b     | A retired person plans to invest up to Rs. 50,000 in NSC certificates yielding 12 percent interest per year and corporate bonds yielding 15 percent interest per year. The person is told to invest a minimum of Rs. 25,000 in the NSC certificates and a minimum of Rs. 10,000 in the corporate bonds, with no more than one-fourth of the total investment to be in corporate bonds. Formulate the given problem as an LPP to decide how the manager should invest in each type of bond to achieve the maximum annual interest. | 2.5   | CO1 |
| 3a     | Solve the following LPP using simplex method and discuss the nature of the solution:<br>Min $Z = -4x_1 + x_2$<br>subject to<br>$x_1 - 2x_2 \leq 2$ ,<br>$-2x_1 + x_2 \leq 2$ ,<br>$x_1, x_2 \geq 0$ .                                                                                                                                                                                                                                                                                                                             | 2.5   | CO1 |
| 3b     | Consider a balanced transportation problem (TP) having 2 sources and 3 destinations such that the availability at each source is 30 units and the demand at each destination is 20 units. Given that the dual variables in the optimal table corresponding to sources and destinations are -1, 2, and 1, 2, 3, respectively. Then, find the optimal objective function value of the given TP.                                                                                                                                     | 2.5   | CO2 |

| 4a    | <p>Consider the LPP</p> <p>Max <math>6x_1 + 12x_2 + 12x_3 - 6x_4</math></p> <p>subject to</p> <p><math>x_1 + x_2 + x_3 = 4,</math></p> <p><math>x_1 + 4x_2 + x_4 = 8,</math></p> <p><math>x_1, x_2, x_3, x_4 \geq 0.</math></p> <p>The optimum table is given by:</p> <table><tr><th>Basis</th><th>Solution</th><th><math>x_1</math></th><th><math>x_2</math></th><th><math>x_3</math></th><th><math>x_4</math></th></tr><tr><td><math>x_3</math></td><td>2</td><td><math>3/4</math></td><td>0</td><td>1</td><td><math>-1/4</math></td></tr><tr><td><math>x_2</math></td><td>2</td><td><math>1/4</math></td><td>1</td><td>0</td><td><math>1/4</math></td></tr><tr><td><math>z</math></td><td>48</td><td>6</td><td>0</td><td>0</td><td>6</td></tr></table> <p>Write the dual problem and find the solution of the dual problem from the optimal solution of primal problem without solving the dual.</p> | Basis | Solution | $x_1$ | $x_2$  | $x_3$ | $x_4$ | $x_3$ | 2  | $3/4$ | 0  | 1     | $-1/4$ | $x_2$ | 2  | $1/4$ | 1     | 0  | $1/4$ | $z$ | 48 | 6     | 0  | 0  | 6  | 2.5 | CO2 |     |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------|-------|--------|-------|-------|-------|----|-------|----|-------|--------|-------|----|-------|-------|----|-------|-----|----|-------|----|----|----|-----|-----|-----|
| Basis | Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $x_1$ | $x_2$    | $x_3$ | $x_4$  |       |       |       |    |       |    |       |        |       |    |       |       |    |       |     |    |       |    |    |    |     |     |     |
| $x_3$ | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $3/4$ | 0        | 1     | $-1/4$ |       |       |       |    |       |    |       |        |       |    |       |       |    |       |     |    |       |    |    |    |     |     |     |
| $x_2$ | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $1/4$ | 1        | 0     | $1/4$  |       |       |       |    |       |    |       |        |       |    |       |       |    |       |     |    |       |    |    |    |     |     |     |
| $z$   | 48                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 6     | 0        | 0     | 6      |       |       |       |    |       |    |       |        |       |    |       |       |    |       |     |    |       |    |    |    |     |     |     |
| 4b    | <p>Solve the following LPP using dual simplex method:</p> <p>Min <math>Z = x_1 + x_2</math></p> <p>subject to</p> <p><math>2x_1 + x_2 \geq 4,</math></p> <p><math>x_1 + 7x_2 \geq 7,</math></p> <p><math>x_1, x_2 \geq 0.</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2.5   | CO2      |       |        |       |       |       |    |       |    |       |        |       |    |       |       |    |       |     |    |       |    |    |    |     |     |     |
| 5a    | <p>Justify the following statements with proper reasoning whether these are correct or incorrect.</p> <p>1. If primal has no feasible solution then dual has always unbounded solution.</p> <p>2. A primal and dual both can have an unbounded optimal solution.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2.5   | CO2      |       |        |       |       |       |    |       |    |       |        |       |    |       |       |    |       |     |    |       |    |    |    |     |     |     |
| 5b    | <p>Consider the problem of assigning 4 operators <math>O_1, O_2, O_3, O_4</math> to 4 machines <math>M_1, M_2, M_3, M_4</math>. The assignment cost (in Rs.) is given in the following table.</p> <table><tr><td></td><td><math>M_1</math></td><td><math>M_2</math></td><td><math>M_3</math></td><td><math>M_4</math></td></tr><tr><td><math>O_1</math></td><td>18</td><td>26</td><td>17</td><td>11</td></tr><tr><td><math>O_2</math></td><td>13</td><td>28</td><td>14</td><td>26</td></tr><tr><td><math>O_3</math></td><td>38</td><td>19</td><td>18</td><td>15</td></tr><tr><td><math>O_4</math></td><td>19</td><td>26</td><td>24</td><td>10</td></tr></table>                                                                                                                                                                                                                                         |       | $M_1$    | $M_2$ | $M_3$  | $M_4$ | $O_1$ | 18    | 26 | 17    | 11 | $O_2$ | 13     | 28    | 14 | 26    | $O_3$ | 38 | 19    | 18  | 15 | $O_4$ | 19 | 26 | 24 | 10  | 2.5 | CO2 |
|       | $M_1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $M_2$ | $M_3$    | $M_4$ |        |       |       |       |    |       |    |       |        |       |    |       |       |    |       |     |    |       |    |    |    |     |     |     |
| $O_1$ | 18                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 26    | 17       | 11    |        |       |       |       |    |       |    |       |        |       |    |       |       |    |       |     |    |       |    |    |    |     |     |     |
| $O_2$ | 13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 28    | 14       | 26    |        |       |       |       |    |       |    |       |        |       |    |       |       |    |       |     |    |       |    |    |    |     |     |     |
| $O_3$ | 38                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 19    | 18       | 15    |        |       |       |       |    |       |    |       |        |       |    |       |       |    |       |     |    |       |    |    |    |     |     |     |
| $O_4$ | 19                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 26    | 24       | 10    |        |       |       |       |    |       |    |       |        |       |    |       |       |    |       |     |    |       |    |    |    |     |     |     |